■ PulmoTwin System Architecture Diagram

Multimodal Cardiopulmonary Digital Twin: Continuous Ingestion, Dynamic Simulation, Safety Rails & Clinical Decision Support

Stage 1: SENSE & INGEST	Stage 2: SIMULATE & TWIN	Stage 3: WHAT-IF REASONING	Stage 4: SAFETY GOVERNANCE	Stage 5: ACT & SUPPORT
1. MULTIMODAL INGESTION **• Longitudinal EHR (ABDM / FHIR R4)** - Baseline Spirometry (FEV1, FVC, PEF) - Biomarkers (FeNO, Blood Eosinophils, IgE) - Documented Allergens & Exacerbation History **• Wearable Telemetry Stream** - Continuous PPG: Heart Rate & HRV (rMSSD) - Blood Oxygen Saturation (SpO2 %) - Respiratory Rate & Skin Temperature - BLE Smart Inhaler Doses & Actuation Log **• Environmental Dispersion API** - Open-Meteo 48h Hourly PM2.5, NO2, SO2, O3 - Thermal Inversion & Weather Telemetry	**2. DIGITAL TWIN SIMULATOR** **• Physiological State Modeling** - Airway Resistance Raw (cmH2O/(L/s)) - Dynamic Lung Compliance Cdyn (mL/cmH2O) - Alveolar-Arterial O2 Gradient (A-a DO2) - Autonomic Sympathetic Tone Index **• 24-Hour Forward Projection** - Differential Equations of Airway Narrowing - Particulate Inhalation Kinetics - Exacerbation Risk Index (ERI) Curve - Estimated Time to Decompensation	**3. COUNTERFACTUAL WHAT-IF** **• Virtual Intervention Sandbox** - Test interventions in silico before execution - Scenario A: Indoor HEPA air filtration - Scenario B: Proactive SMART inhaler puff - Scenario C: Activity rescheduling **• Comparative Efficacy** - ΔERI Peak Risk Reduction % - ΔPEF Airway Improvement - Optimal Personalized Protocol	**4. SAFETY & GOVERNANCE** **• Deterministic Safety Rails** - Dual-layer: Regex & GINA Step-5 boundaries - Red-Flag Distress Interceptor (Bypasses LLM) - Output Sanitizer: Blocks dosage changes - Bilingual Emergency 108/112 SOS Protocols **• Human-in-the-Loop Approval** - Zero-Trust External Action Queue - Gated Caregiver SMS / n8n Webhook - Tamper-evident SQLite Audit Log	**5. ACTION & INTERACTION** **• Patient Experience (Web UI)** - Editorial Design System (Warm Ivory & Glass) - Bilingual English / Hindi Voice & Text - 24h Safest Clean Air Window Finder **• Clinician Decision Support** - Instant 30-Day GINA Summary PDF - Reliever puff frequency & PEF dynamics - GDPR / DPDP Full Data Portability